POWER

VBAT SENSING

VBATT
R24 36K
U5A LM358_DFN
PDM_ADC_IN0
R25 4.7K
GND

3.3V

+LDIN
PWR_FLAG
U6 AMS1117-3.3
VI VO GND
+3.3V
C21 10u
C22 0.1u
0.1u C34
10u C41
GND

TP13 TestPoint
D3 PMEG3020EJ
VBATT
C38 10u
C42 10u
C43 10u
U7 AP62250WU-7
VIN EN BST SW FB GND
L2 4.7uH
C44 330n
+5V
R27 82.5K
R28 100K
C45 22u
C46 22u
D2 POWER_LED
R50 330
D5 PMEG3020EJ
+LDIN
VBATT
P N SH1 SHIELD
J3 XT60PW-M
GND
LI-PO 3S-4S TO 5V

DECOUPLING CAPASITOR

+3.3V
C18 2.2u
C17 2.2u
C16 0.1u
C15 0.1u
C14 0.1u
C19 0.1u
C20 0.1u
GND

CURRENT SENSOR

U4 ACS758xCB-100B-PSF
VIOUT VCC GND IP+ IP-
VBATT
IP-
R30 150
IC2 BTS722204ESAXUMA1
GND_1 VDD SO SI SCLK CSN LHI IN0 IN1 IN2 IN3 IS VS
OUT0_4 OUT0_3 OUT0_2 OUT0_1 OUT1_2 OUT1_1 OUT2_2 OUT2_1 OUT3_4 OUT3_3 OUT3_2 OUT3_1
SPI1_MISO
SPI1_MOSI
SPI1_SCLK
SPI1_CS
CURRENT_SENSE_1
CURRENT_SENSE_2
CURRENT_SENSE_3
CURRENT_SENSE_4
PDM_ADC_IN2
4.7K R29
C25 10n
R26 4.32K
CURRENT_OUT0
CURRENT_OUT1
CURRENT_OUT2
CURRENT_OUT3
C26 10n
C27 10n
C28 10n
C29 10n
GND

TJA1051T/3

+5V
PWR_FLAG
VCC
C2 0.1u
IC3 TJA1051TK_3.118
TXD GND VCC RXD VIO CANL CANH S EP
CAN1_TX
CAN1_RX
+3.3V
R33 120
O_CAN2L
O_CAN2H
TVS DIODE
IC4 PESD1CAN-U
GND

MCU

NRST
BOOT0
OSC_IN
OSC_OUT
Y1 8MHz
C23 18p
C24 18p
R1 10K
R2 10K
C1 0.1u
C3 2.2u
C4 2.2u
PD2
PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7 PC8 PC9 PC10 PC11 PC12 PC13 PC14 PC15
VCAP_1 VCAP_2
VBAT VDD VDD VDD VDD VDDA
VSS VSSA
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB9 PB10 PB11 PB12 PB13 PB14 PB15
PDM_ADC_IN0
PDM_ADC_IN2
PDM_ADC_IN3
SPI1_SCLK
SPI1_MISO
SPI1_MOSI
CAN1_RX
CAN1_TX
SWDIO
SWCLK
CURRENT_SENSE_3
CURRENT_SENSE_2
CURRENT_SENSE_1
MUX_INPUT_A0
MUX_INPUT_A1
MUX_INPUT_A2
CAN1_EN
CAN1_STB
CAN1_ERR
I2C1_SDA
I2C1_SCL
U1 STM32F405RGTx
PWR_FLAG
GND

INPUT_0 … INPUT_7

INPUT_0 TP1 SP
INPUT_1 TP2 SP
INPUT_2 TP3 SP
INPUT_3 TP4 SP
TP5 SP
TP6 SP
+3.3V
SP TP12 INPUT_4
SP TP11 INPUT_5
SP TP10 INPUT_6
SP TP9 INPUT_7
SP TP8
SP TP7
+5V
GND

CURRENT_OUT0 … CURRENT_OUT3

J1 Screw_Terminal_01x04
GND

I2C1_SDA / I2C1_SCL

I2C1_SDA TP14 SP
I2C1_SCL TP15 SP
VBATT TP16 SP
+3.3V
R32 4.7K
R31 4.7K
I2C1_SDA
I2C1_SCL

O_CAN2H / O_CAN2L

O_CAN2H
O_CAN2L
J4 Screw_Terminal_01x03
GND

SWDIO / SWCLK / NRST

+3.3V
SWDIO
SWCLK
NRST
J2 B5B-XH-A
GND

U3 CD74HC4067M

+3.3V
C13 10u
GND
PDM_ADC_IN3
MUX_INPUT_A0
MUX_INPUT_A1
MUX_INPUT_A2
COM S0 S1 S2 S3 E VCC GND
I0 I1 I2 I3 I4 I5 I6 I7 I8 I9 I10 I11 I12 I13 I14 I15
INPUT_PIN0 INPUT_PIN1 INPUT_PIN2 INPUT_PIN3 INPUT_PIN4 INPUT_PIN5 INPUT_PIN6 INPUT_PIN7
GND

D1 SP720ABTG

+3.3V
INPUT_PIN0 INPUT_PIN1 INPUT_PIN2 INPUT_PIN3 INPUT_PIN4 INPUT_PIN5 INPUT_PIN6
IN_1 IN_2 IN_3 IN_4 IN_5 IN_6 IN_7 V-
V+ IN_14 IN_13 IN_12 IN_11 IN_10 IN_9 IN_8
INPUT_PIN7
GND

VINPUT SENSING

INPUT_PIN6 INPUT_6 R21 38K R23 4.7K C12 10u
INPUT_PIN0 INPUT_0 R9 38K R8 4.7K C5 10u
INPUT_PIN1 INPUT_1 R14 38K R15 4.7K C8 10u
INPUT_PIN7 INPUT_7 R20 38K R22 4.7K C11 10u
INPUT_PIN2 INPUT_2 R10 38K R11 4.7K C6 10u
INPUT_PIN3 INPUT_3 R12 38K R13 4.7K C7 10u
INPUT_PIN4 INPUT_4 R16 38K R17 4.7K C9 10u
INPUT_PIN5 INPUT_5 R18 38K R19 4.7K C10 10u
GND

MOUNTING HOLES

H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
GND

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File: PDM.kicad_sch
Title:
Size: A2 | Date: | **Rev:**
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